

Appendix

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(APPENDIX) Appendix

1 Appendix

1.1 Alternative Response and Further Control Variables

Though this paper only uses $\% \Delta \text{Consumption}_d$ as a response variable,¹ it is vulnerable to regional variations in inflation and economic shocks. While the former can be partially controlled for using state-specific CPI data from the Central Statistics Office, possible impacts of and controls for the latter still need to be researched.

In the meantime, other response variable such as $\% \Delta \text{Wages}_d$, the percent change in district d 's average household wages from 2007-08 to 2017-18, sourced from the “Employment, Unemployment, and Migration” survey of 2007-08, another part of the 64th Round of the NSS ([National Sample Survey Office 2010](#)), as well as from the Periodic Labor Force Survey (PLFS) of 2017-18 ([National Statistical Office 2019](#)), may be informative. On one hand, the stickiness of wages may control for potential volatility in $\% \Delta \text{Consumption}_d$. On the other hand, wages may not be affected by the lifelong consumption smoothing underlying $\% \Delta \text{Consumption}_d$ —in other words, wages be less responsive to unobserved shocks while still being *more* responsive to any effect of EMIE_d which may manifest in the decade-long window we study. None of this diminishes the fact that informal, self-employed, and otherwise non-wage laborers are very common in India. Individuals vary in the degree to which wages reflect their true income, and this variation may not be exogenous to our model. Numerous papers studying similar contexts in India have still used wages as their primary response variable ([Shastry 2012](#); [Azam et al. 2013](#); [Madheswaran and Singhari 2016](#); [Refeque and Azad 2022](#)). $\% \Delta \text{Employment}_d$, the percent change in district d 's employment, sourced from National Sample Survey Office ([2010](#)) and National Statistical Office ([2019](#)), may provide additional insights as well.

¹Which itself is a flawed measure due to the spurious correlations which may arise from regressing on a ratio ([Kronmal 1993](#)). Future work will either replace this with Kronmal's proposed model (akin to “multiplying” the response variable's denominator across the right-hand side) or with a simple log difference.

$\Delta\text{Gini}_d^{\text{Consumption}}$ and $\Delta\text{Gini}_d^{\text{Wages}}$, the change in the Gini coefficient of district d 's consumption and wages respectively from 2007-08 to 2017-18, would share the same sources as their non-Gini counterparts above. The models of Equation [?@eq-iv-eq](#) were actually estimated using $\Delta\text{Gini}_d^{\text{Consumption}}$ as the response variable; this returned a near-zero R^2 and adjusted R^2 , meaning the model did no better than the sample mean. Within-district distributional effects of EMIE, if they exist, almost certainly need more than 10 years to manifest.

$\%\Delta\text{HDI}_d$, the percent change in district d 's Human Development Index (HDI), sourced from the National Family Health Surveys of 2005-06 (NFHS-III) and 2019-21 (NFHS-V), would provide a more holistic, shock-resistant measure of development, even if the developmental effects of EMI may need more than 15 years to reveal themselves and even if better alternatives (e.g., the HDI_a of [Chaurasia 2023](#)) exist. But NFHS data has unfortunately been unavailable since the USAID funding cuts of 2025.

To further control for omitted variables and to further reveal the mechanisms behind EMIE's effect on consumption, future work may include TechFirms_d , a measure of the number of IT firms from the 2005 Economic Census ([Central Statistics Office 2008](#));² infrastructure measures beyond the proportion of pucca (i.e., permanent) housing; and proxies for corruption, chronic teacher absenteeism, etc., all of which are assumed to be similar across districts. Shastri (2012) incorporates corruption by including a "Hindi Belt" indicator for states such as Bihar, Delhi, Uttar Pradesh, and Rajasthan, all majority Hindi-speaking states known for their "high levels of corruption and government inefficiency" (p. 299). Note that a desire for EMI *and* a desire for better teacher quality among poorer households is what drove a 30% increase in the number of private schools from 2012 to 2020, most of them low-cost private schools ([Gooptu and Mukherjee 2023, 2](#)). This did not seem to change the fact that private schools attract richer households than government schools on average ([Muralidharan and Sundararaman 2015](#)), but it does imply a relationship between cost, private vs. government schools, and teacher quality which may inform future proxies of teacher quality. Adding in a control for migration flows will be key as well, to ensure that the effect of EMIE does not dissipate above a certain threshold due to all the out-migration to urban hubs it has facilitated. See Sec. [Section 1.2.2](#) for more on migration and spatial effects.

Interaction terms such as $\text{EMIE}_d \times \text{Urban}_d$ and $\text{EMIE}_d \times \text{TechFirms}_d$ may also be key in identifying the mechanisms by which EMIE can affect a district's growth. Even if households have identical underlying preferences for how to educate their children, the menu of choices available to them (that is, the feasible set over which they optimize) may still be affected by factors far larger than their own education-related decision-making—whether that be by the presence of IT firms nearby, or by the parents' familiarity with and ability to learn more about the Indian education system. To better understand the meaning of the

²State fixed effects should absorb many predictors of this e.g., labor regulation ([Besley and Burgess 2004](#)).

highly significant coefficient on urbanization in Table [?@tbl-cons-iv](#), the urbanization and EMI interaction may be complemented with an urbanization and parental/household head education interaction. The low Muslim/ST/OBC estimates, too, can be studied further by interacting them baseline consumption shares, urbanization, and EMIE itself. And finally, to understand the small coefficients and relative insignificance of the household head’s education estimates, we could interact these education levels with urbanization and baseline consumption, or we could even try replacing our baselines with changes, as with a first-difference estimator.

Notice how all actual and proposed regressors in this model reflect baseline exposures. Replacing these with changes and time-varying instruments instead may allow us to use a Bartik-style methodology (e.g., [Goldsmith-Pinkham et al. 2020](#)). Linguistic distance, for example, could be turned into a time-varying instrument if interacted by a nation or state-wide shock in EMI supply or demand between 2007-08 and 2017-18. We have yet to identify shocks which have both a plausible effect on EMIE (e.g., changing curricular standards or waves of English training for teachers) and which vary across time alone, not district.

1.2 District Matching and Spatial Autocorrelation

1.2.1 District Matching Method

There are three reasons that district labels vary between 2001 to 2007-08 to 2017-18 in the data: genuine political changes, variation in anglicization, and typos. Fuzzy joins (described below) were used to correct for the latter two.

Political changes, however, were not so easily addressed. The vast majority of such changes were one of five kinds: name changes, clean partitions (one district is partitioned into multiple), clean mergers (multiple districts merge into one), carve-outs (*pieces* of multiple districts combine into one), and boundary shifts ([India State Stories 2024](#)). Without knowing precisely where in its district each household was at the time of sampling, it is not feasible to match districts across carve-outs and boundary shifts. Our district matching method is thus reliant on the assumption that all district changes were either name changes, clean partitions, or clean mergers.

This seems to be precisely the same assumption made by [India State Stories \(2024\)](#) and [Jaacks et al. \(2020\)](#), whose data, despite containing some factual errors and typos, was the basis of our district matching method. To justify this assumption we turn to [Kumar and Somanathan \(2016\)](#), who up until 2001 are able to track the *proportion* of each district’s population that was allocated into a new district as a result of district changes. They label these changes as one of two types: clean partitions and non-partitions (which would include name changes, clean mergers, carve-outs, and border shifts). By extracting their results using Tabula ([Aristaran et al. 2018](#)), we know that 93.4% of non-partitions were either name changes or transfers of uninhabited land ([Kumar](#)

and Somanathan 2016). These district changes are plotted in Figure ?@fig-districtcarveoutsshifts-fig.

We thus assume no district changes were carve-outs or border shifts. To match districts, we ran a hierarchical, cascading sequence of iterated fuzzy joins. Data from 2001, 2007-08, and 2017-18 is first linked to the closest year in the district trackers of India State Stories (2024) and Jaacks et al. (2020). Districts are then matched if they meet a certain string/phonetic distance threshold. Matches were made in the following order:

1. **soundex** = 0, for phonetically equivalent variations in anglicization (e.g., “Baleswar” and “Balasore”).
2. **qgram** distance = 0, for rearrangements of bigrams (e.g., “East Godavari” and “Godavari East”).
3. Jaro-Winkler distance ≤ 0.15 , for respellings and vowel swaps with 0.85 degrees of similarity.
4. Full Damerau-Levenshtein distance ≤ 2 , for ≤ 2 insertions, deletions, substitutions, and transpositions.
5. Optimal String Alignment distance ≤ 1 , for a single typo.

Any district from the 2001, 2007-08, or 2017-18 data which remains unmatched after this process is then matched with the next closest year in the district tracker. This process repeats until all years in the tracker had been exhausted. 80 still-unmatched districts were then manually matched.

The current method, however, attempted to mirror the mergers and partitions which characterize district changes by using many-to-many matching for 2001 to 2007-08 and from 2007-08 to 2017-18. The number of duplicate values this creates, particularly for 2001 and 2007-08 measures, seems to lead to massive multicollinearity when state fixed effects are added (see Sec. ?@sec-iv-iv). Better data harmonization is essential: future work will merge 2017-18 survey-weighted estimates into 2007-08 districts using population weights from the 2011 census, and will do the same for 2007-08 estimates and 2001 districts using the 2001 census. Our units of analysis will thus become districts in 2001 as opposed to districts in 2017-18. If this solves the rank wreckage likely created by the current district tracking method, then we will be able to include the state fixed effects so important for our instrument in Sec. ?@sec-iv-iv.

1.2.2 Spatial Autocorrelation, Spatial Spillovers, and Migration

As was evident from the maps of Figures ?@fig-map1-fig and ?@fig-map2-fig, numerous districts are still missing from the data. Perhaps the assumption of no carveouts and border shifts was a false one to make. And yet, even if all districts were perfectly matched, problems would arise. The aforementioned figures were constructed using 2019-20 shapefiles from Bhatia (2020), despite our “treatment” year being 2007-08. The splintering of districts into multiple

neighbors over time allocates the same value of the 2007-08 treatment across neighbors, a rank wreckage equivalent to spatial autocorrelation in treatment when using 2019-20 geometry.

The presence of true spatial autocorrelation in this environment is plausible. The spatial spillovers of development and the agglomeration of tech firms are key forms of autocorrelation which render the assumption of identical and independent distributions false, and thus many of the results of OLS meaningless. But special attention must be paid to the role of migration here. Note that, if part of EMI’s effect is in enabling children to migrate across the country (e.g., to districts which have more tech firms), then the true structural parameter $EMIE_d$ in our model would only reflect the EMI-induced benefits which they send back e.g., as remittances. In other words, even in a world of perfect shapefiles, our results could show that EMIE is a substandard tool for local area’s development even if it is actually a great tool for a local population’s development. How, then, could the effect of EMI ever be identified without being dissipated away by between-district migration?

Chakraborty and Bakshi (2016) find a 4% decadal rate of inter-district migration in their states of interest (West Bengal, Haryana, and Punjab) up until 2001, a migration rate³ so low that they assume migration away almost entirely, effectively equating the district where one’s education occurred with the district most affected by their education.⁴ Topalova (2005) find decadal inter-district migration rates to be around 3.2% and 13.1% of rural and urban areas respectively, or around 0.5% and 4.0% when only considering migration related to employment, trends which were “remarkably constant” across her years of study (1983, 1987, and 1999), including “no visible increase after the economic reforms of 1991” (p. 24), which preceded to the IT growth and service job offshoring described in Sec. ?@sec-intro; National Statistical Office (2021) imply not much has changed since either, albeit using 2020-21 data. It seems to be a structural, secular trend in India that migration only occurs over short distances.⁵

³We define an area’s ‘migration rate’ over a period as the share of this area’s end-of-period population who report having migrated over said period. The ‘decadal inter-district migration rate for rural areas in 2001’, for example, would refer to the “percent of people living in rural areas [who in 2001] reported changing [their district] within the past 10 years” (Topalova 2005, 25). Social scientists only look at migrations reported within the past one, five, ten, etc. years to both capture shorter-term fluctuations in migration trends and to avoid the cumulative nature of metrics built on all lifetime migrants (Mahapatro 2020).

⁴More precisely, they “assume that district of current residence (or of employment) of an individual is approximately the same as the schooling district” (Chakraborty and Bakshi 2016, 6).

⁵And perhaps only for short times as well, though this may only be the product of data interpretation errors. Keshri and Bhagat (2013), for example, use 2007-08 data from the National Sample Survey Office (2010) to find a temporary (1-6 months) labor migration rate of 26% in rural areas, 13 times larger than the permanent labor migration rate. And yet, despite using the same 2007-08 data, Lucas (2021b) finds this same rate to be 2.6%. Barring the possibility of a misplaced decimal point, the sole difference seems to be Lucas’s observation that the NSS Office’s definition of ‘temporary’ migration only considered if a migration was *intended* to be temporary, not if it actually was; he adjusts his measures accordingly, whereas Keshri and Bhagat, it seems, do not. Imbert and Papp (2020) corroborate Lucas’s number.

Why could this be the case? Topalova (2005) identifies marriage as the largest factor behind migration writ large, but this effect is driven entirely by women:⁶ using 2020-21 data, the National Statistical Office (2021) demonstrate that male migration is instead driven almost entirely by economic factors. For example, Kochar (1999) finds that after negative farm income shocks, any consumption loss households would face is offset by men taking on more off-farm work, implying that local labor markets in rural India are sufficiently diversified, local social capital is sufficiently anchoring, and other markets are sufficiently segmented off for households to consumption smooth using at most short-distance, temporary migration. “Households use migration as an ex post income-smoothing mechanism,” states Morten (2019, 3), whose structural model on the interplay of risk sharing and migration (particularly rural-urban migration) corroborates the role of risk in limiting such migration just as Munshi and Rosenzweig (2016) corroborate the roles of risk and social capital; the latter authors find caste-based networks of rural insurance significantly constrain rural males’ ability to migrate, and structural estimation indicates a large effect on migration from marginal improvements in formal insurance, particularly on rural-urban migration. Administrative differences may also play a factor, though Kone et al. (2018) imply that such differences matter far more across states’ borders than districts’; by using 2001 census data, in fact, they find rates of long-distance inter-*state* migration to be far lower than those of Brazil and China, which they posit is the product of non-portable state-level welfare schemes plus states’ preference to hire their own residents in the public sector.

Newer findings, however, repudiate many of these low migration rates wholesale. Lucas (2021a) finds that, using 2007-08 data from the National Sample Survey Office (2010), the 4.7% net flow of migrants from rural to urban areas in 2003-08 was among the highest of all countries studied.⁷ More novel methods, however, were used by the Economic Division (2017). Their cohort-based migration metric⁸ finds the inter-district migrant share alone to be 6.6% when applied to 2001 and 2011 census data. Railway passenger data is then used to proxy for labor-related migration flows which are found to extend well beyond even state boundaries, almost entirely between large urban hubs. The final finding relevant to us, shared by both the Economic Division (2017) and the more standard analysis of Mahapatro (2020), is that the stagnant migration rates of the late 20th century, as observed by Topalova (2005), have increased since 2001, perhaps the product of economic growth which has raised the benefits of

⁶Though a large majority of women list marriage as their primary reason for migrating, over the past few decades economic factors have gained greater and greater explanatory power (Mahapatro 2020), and women who list marriage will increasingly begin working right after migrating anyways (Rajan et al. 2020). The relationship between social capital, information networks, and economically-motivated female migration is further explored by Neetha (2004).

⁷Lucas (2021b) also uses NSS data to find far, far smaller rates of short-term migration, with 2.6% of rural adults migrating for 1-6 months versus the 50% estimates cited by, say, Morten (2019).

⁸Which defines net migration as “the percentage change in population between the 10-19 year-old cohort in an initial census period and the 20-29 year-old cohort in the same area a decade later, after correcting for mortality effects” (Economic Division 2017, 267).

migration increasingly above and beyond its costs (Economic Division 2017).

The significance of this new literature for us is two-fold: one, it means the mechanism behind part of Shastry (2012)’s exclusion restriction demonstration in Sec. 4 may not apply as much to our period of interest; and two, it means that EMIE’s effect dissipating away with migration is a genuine possibility. If we tentatively assume that regressors such as urbanization adequately capture the effect of longer-distance migration (which accords with the Economic Division 2017’s finding that it is “rich metropolises [attracting] large inward flows of labour,” p. 277) then the most relevant kind of dissipation for us will be over neighboring district lines (which also accords with Economic Division 2017, 276).⁹

We can control for this spatial autocorrelation by incorporating spatial lags into our model. To test for spatial autocorrelation we would follow p. 323 of Anselin (2001) and construct a Moran’s I statistic; but proper estimation of it would depend on us having proper shapefiles for the year of interest with which to build contiguity neighbor lists. If our unit of analysis ends up being 2001 districts, then this would require 2001 shapefiles. As it stands however, using 2019-20 shapefiles, the p -value for spatial autocorrelation in the residuals of our 2SLS model is `r_m_cons_resid$p.value %>% signif(3)` and for consumption growth `r_m_cons$p.value %>% signif(3)`.¹⁰ Future work will combine shapefiles from the 2011 and 2001 censuses with the data harmonization changes described in Section 1.2.1 to ensure these reflect genuine spatial autocorrelation as opposed to the flaws of our current district tracking methodology.

If spatial autocorrelation is ultimately found to be significant, then the methods of Murgante and Borruso (2012), Pebesma and Bivand (2025), Xu and Wooldridge (2022), and Zhang et al. (2024) would be used to develop a spatial Durbin-type model (a spatial error model, a spatial lag model, a spatial Durbin model, a general nested model, etc., depending on the results of LM tests on the common factor specifications which distinguish these kinds of models), a class of model which incorporates spatial lags of spatially autocorrelated variables as further regressors or in the error. Conley standard errors and/or Kelejian and Prucha (2007) could also be used to create heteroskedasticity and autocorrelation consistent SEs.

2 Technical Note for Replication

Those replicating the code for this paper are encouraged to skim the code chunk titled `Define functions to read in short 8.3 filenames`. There the functions `read_sav_short()`, `read_csv_short()`, and `read_excel_short()` are de-

⁹Reminiscent of Kone et al. (2018), the Economic Division (2017) finds that within-state flows of labor migration are much higher than between-state flows. State fixed effects will be essential for further analysis.

¹⁰Results reflect queen contiguity but are robust to rook contiguity.

fined; if the replicator runs into issues while reading in files, troubleshooting methods are explained there.

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